

YOLO Model Testing Report

Model:	yolov10n
Model Size:	5.6 MB
Parameters:	2.8 M
FLOPs (640x640):	8.74 GFLOPs
Run Name:	yolov10n_testing_20251124_003508
W&B Run Name:	yolov10n_bdd100k_yolo_test_20251124_003508
Timestamp:	2025-11-24 00:38:36
Dataset:	bdd100k_yolo - test split
Images Processed:	20000
Total Execution Time	184.10s
IoU Threshold:	0.5

Inference Performance

Metric	Value
Average Inference Time	3.98 ms per image
FPS (Frames Per Second)	251.30

Note: Inference time includes preprocessing, model inference, and postprocessing.

Overall Accuracy Metrics

Metric	Value
Precision	0.2001
Recall	0.6432
F1-Score	0.3052
mAP@0.5	0.0764

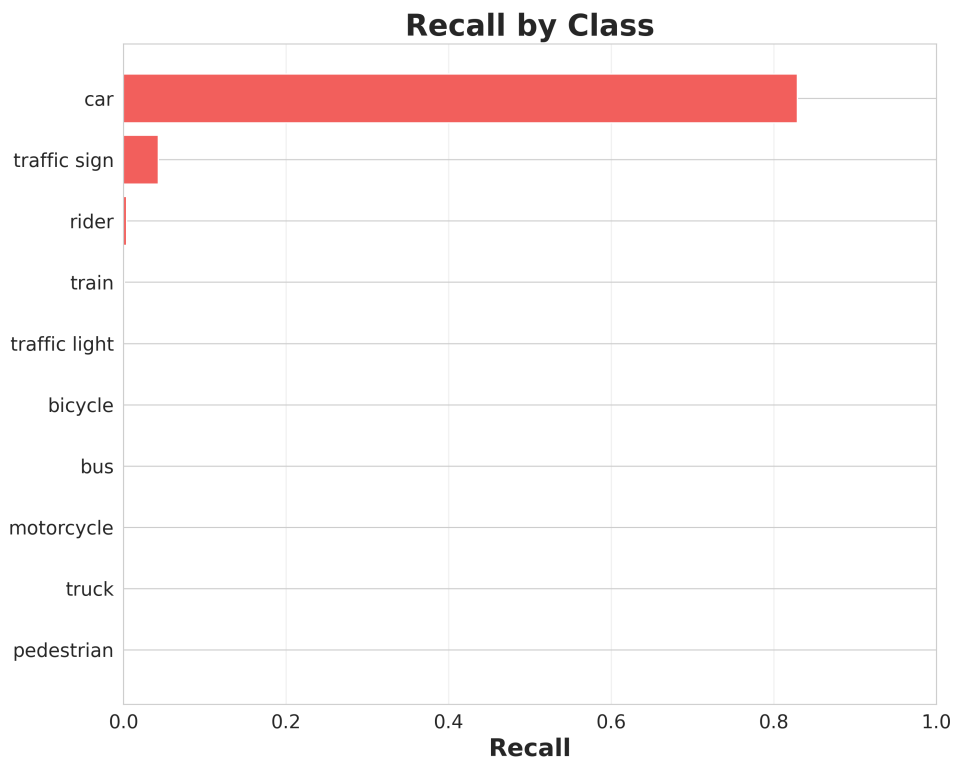
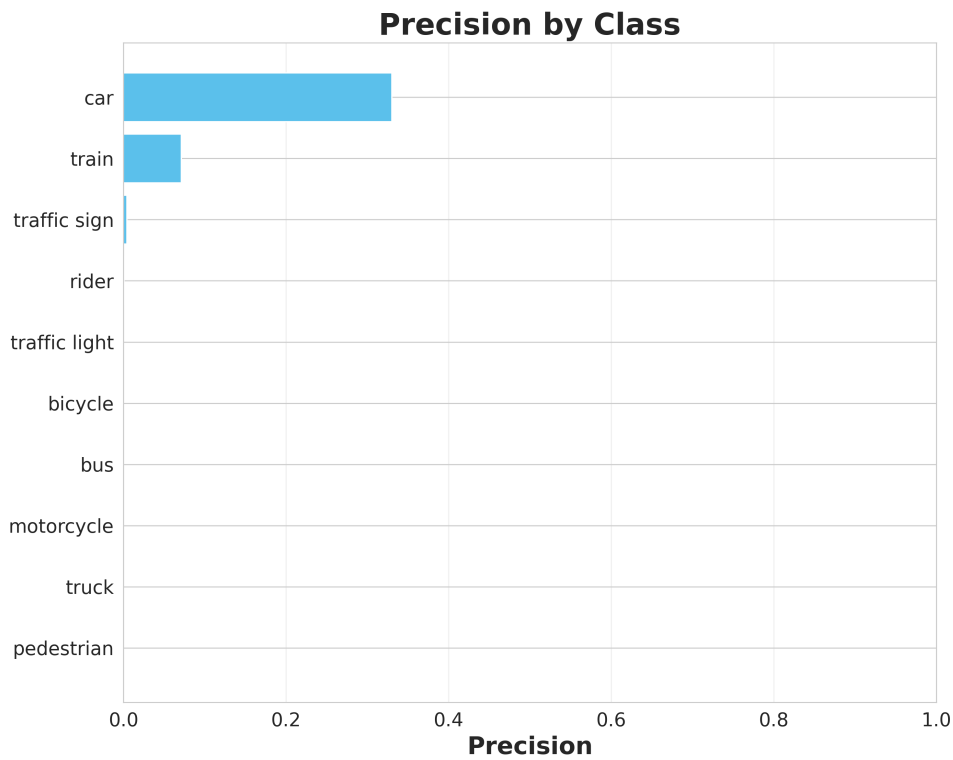
mAP@0.5:0.95	0.0437
True Positives	68055
False Positives	272086
False Negatives	37752

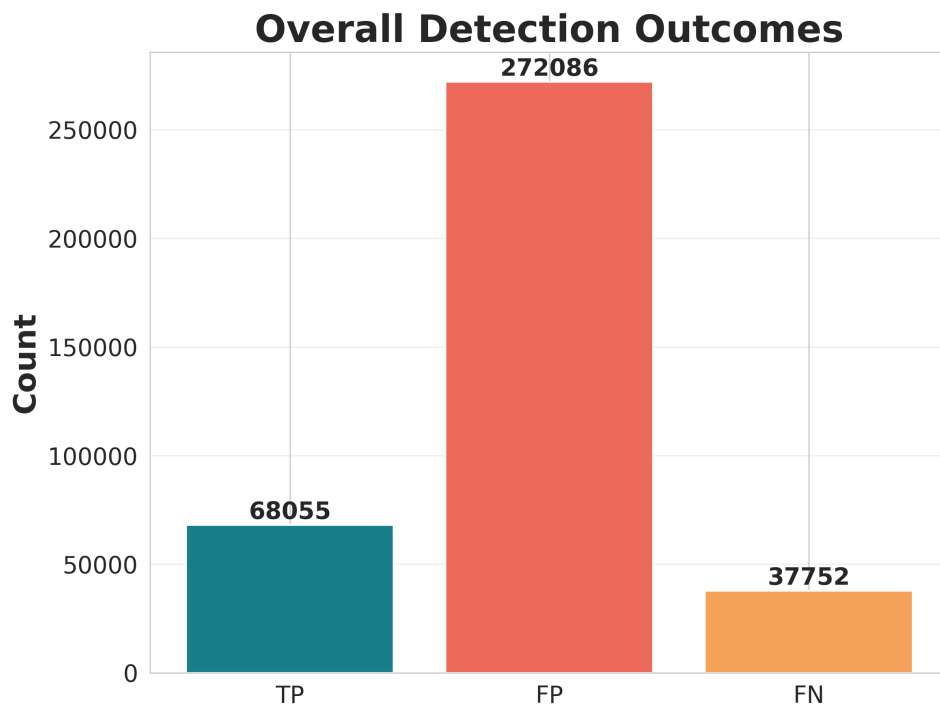
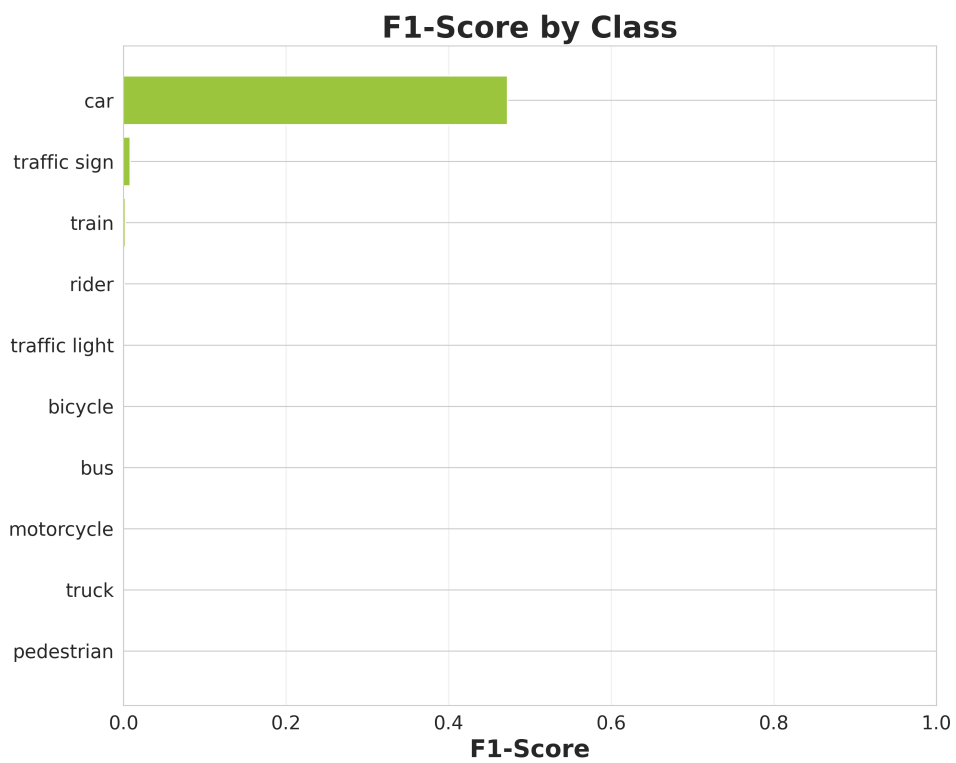
Confusion Matrix

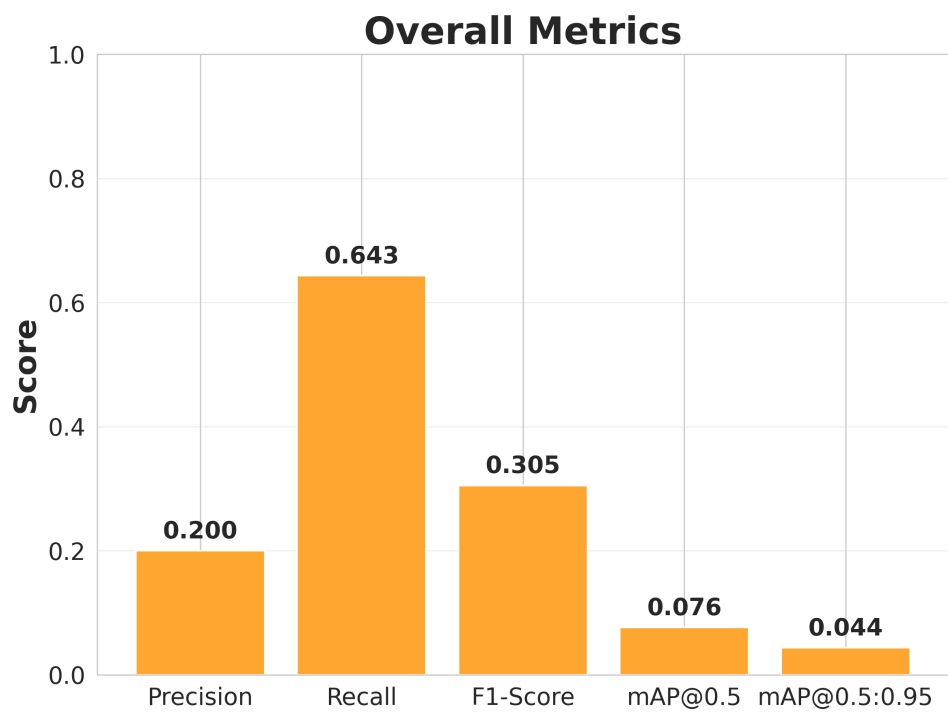
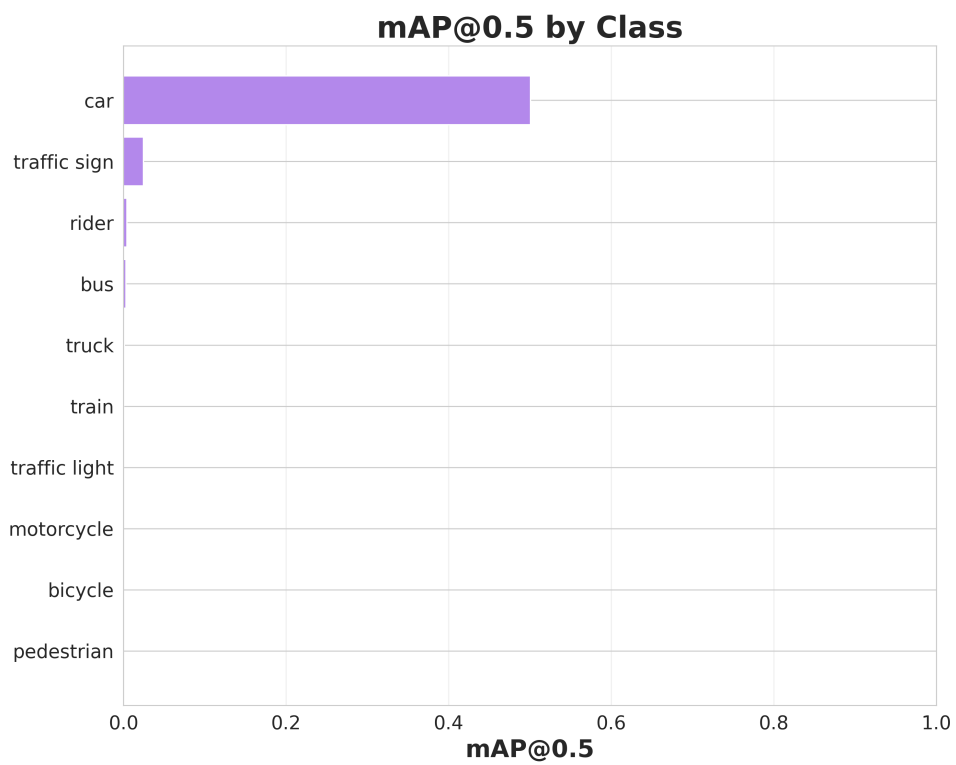
Confusion Matrix (yolov10n validation)										
True Class	pedestrian	380	67	5					3	18
	rider	1								
	car	9	67740	966	166				16	37
	truck	9	8							
	bus			1						
	train		339	258	765	2				
	motorcycle		6	10	3	2				
	bicycle	1	2451	1946	298	1			1	8
	traffic light			1						
	traffic sign			1	1		1			4349
		Predicted Class								
	pedestrian	rider	car	truck	bus	train	motorcycle	bicycle	traffic light	traffic sign

Correct Predictions (Diagonal Sum): 68055
Total Matched Predictions: 366185

Performance Visualizations







Per-Class Performance

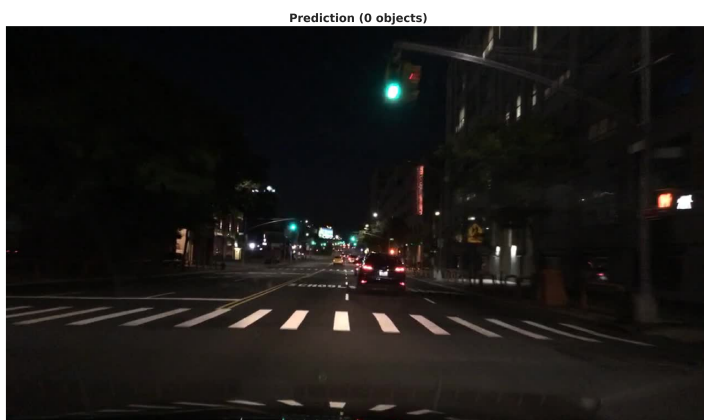
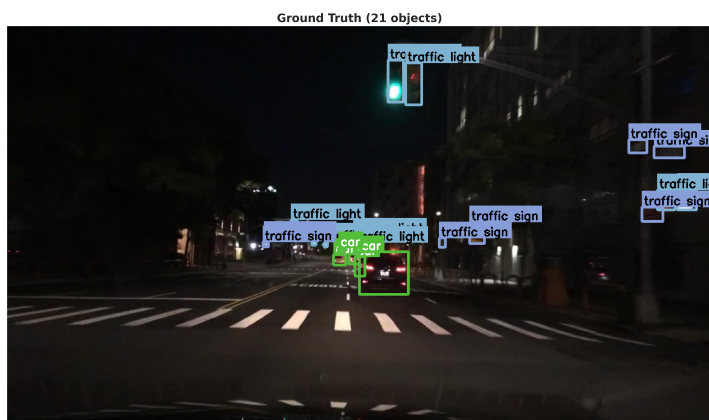
Class	TP	FP	FN	Precision	Recall	F1-Score	mAP@0.5
pedestrian	0	0	8405	0.0000	0.0000	0.0000	0.0000
rider	1	1292	241	0.0008	0.0041	0.0013	0.0045
car	67740	137353	13948	0.3303	0.8293	0.4724	0.5009
truck	0	8701	210	0.0000	0.0000	0.0000	0.0013
bus	0	3217	10	0.0000	0.0000	0.0000	0.0033
train	2	26	1789	0.0714	0.0011	0.0022	0.0002
motorcycle	0	0	72	0.0000	0.0000	0.0000	0.0000
bicycle	0	0	6151	0.0000	0.0000	0.0000	0.0000
traffic light	0	52817	3	0.0000	0.0000	0.0000	0.0001
traffic sign	312	68680	6923	0.0045	0.0431	0.0082	0.0246

Sample Predictions: Ground Truth vs Model

Sample #1: f22976ab-fbf2aee6.jpg

Ground Truth Objects: 21 | Predicted Objects: 0

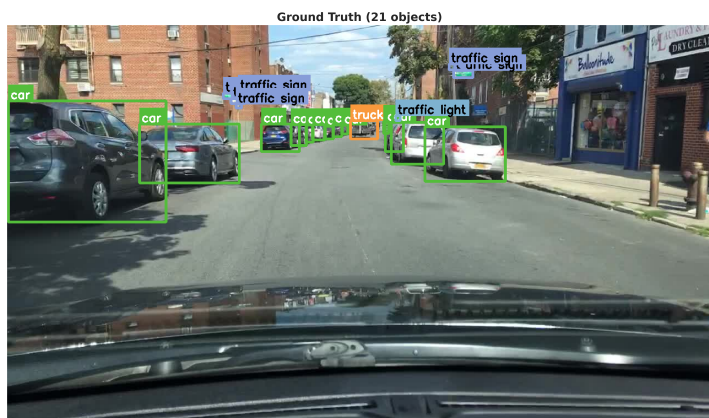
Weather: clear | Scene: city street | Time of Day: night



Sample #2: d2f7022b-fb6cd551.jpg

Ground Truth Objects: 21 | Predicted Objects: 7

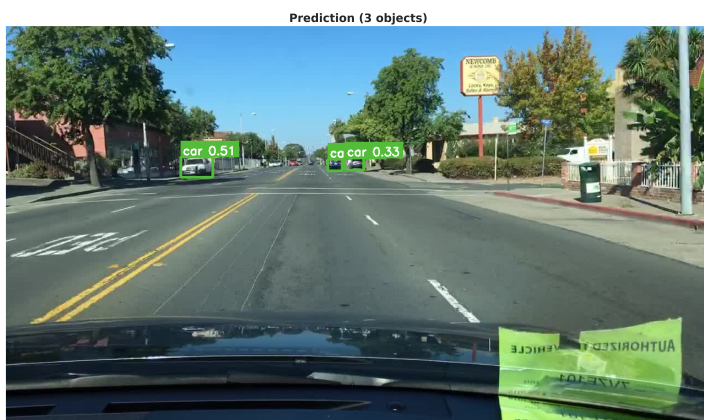
Weather: partly cloudy | Scene: city street | Time of Day: daytime



Sample #3: f11d3f98-4e8ac97b.jpg

Ground Truth Objects: 11 | Predicted Objects: 3

Weather: clear | Scene: city street | Time of Day: daytime



Sample #4: f0600889-d0d0a21b.jpg

Ground Truth Objects: 10 | Predicted Objects: 0

Weather: rainy | Scene: city street | Time of Day: night

Ground Truth (10 objects)



Prediction (0 objects)

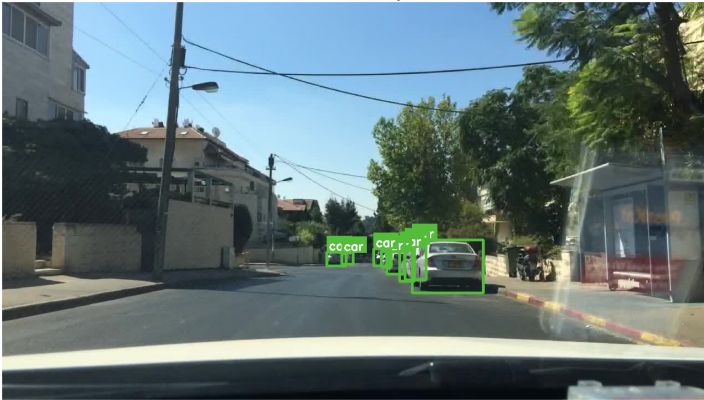


Sample #5: f0fc8a4c-06486914.jpg

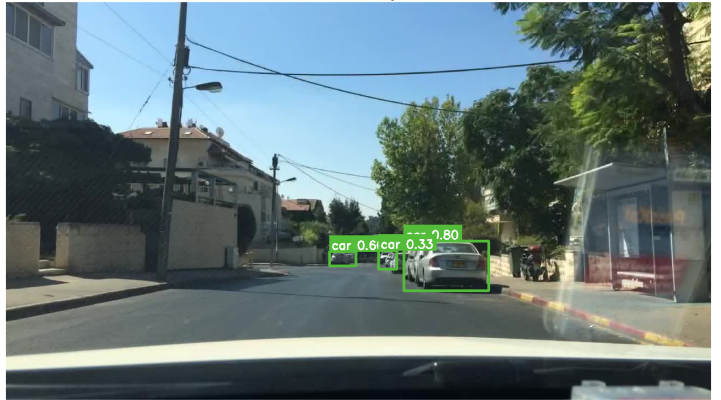
Ground Truth Objects: 8 | Predicted Objects: 3

Weather: clear | Scene: residential | Time of Day: daytime

Ground Truth (8 objects)



Prediction (3 objects)

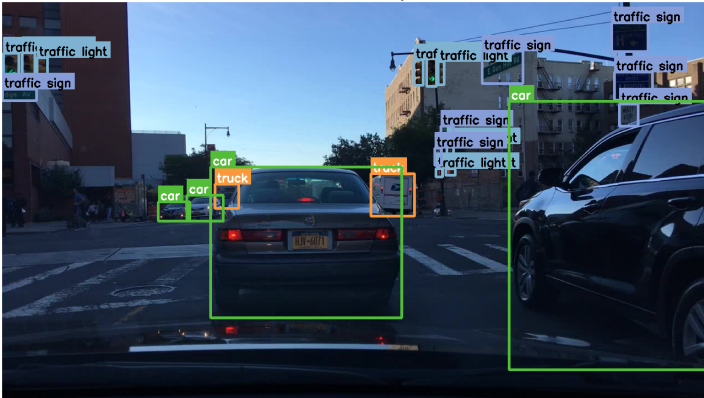


Sample #6: fd35294b-3ea90505.jpg

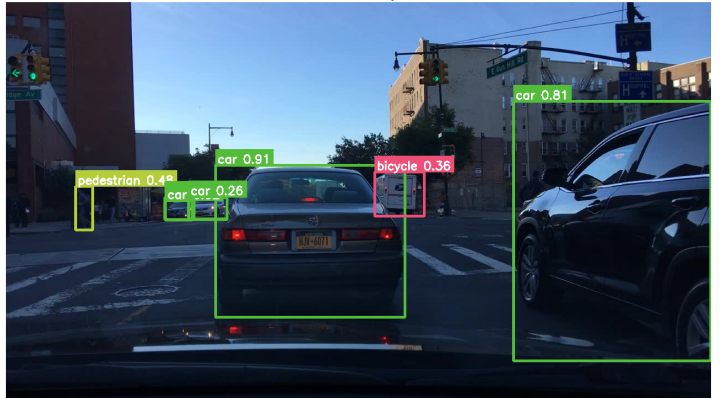
Ground Truth Objects: 22 | Predicted Objects: 8

Weather: clear | Scene: city street | Time of Day: daytime

Ground Truth (22 objects)



Prediction (8 objects)



Generated by YOLO Quick Test Script